



İZMİR INSTITUTE OF TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING

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DESIGN OF A METHANOL-WATER-GLYCEROL RECOVERY SYSTEM

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1. Introduction

The biodiesel processing section addressed in this project concerns the recovery of methanol, water and glycerol from a three-component feed stream with a capacity of 100 kmol/h. According to the project specifications, the separation system should only include two flash distillation units and one continuous multi-stage distillation column. As a result of the recovery process, each of the recovered methanol, water and glycerol products must achieve a purity of at least 95 per cent by molar mass. The report must also include mass and energy balances, operating pressures and temperatures, safety considerations, a justified process configuration, cost items and cost factors, purity and recovery calculations, and possible reuse applications for the recovered streams. [1]

It is assumed that the feed composition consists of 12.92 mol % methanol, 61.18 mol % water and 25.90 mol % glycerol. The selected design strategy makes use of a flash section to remove methanol and water from the glycerol-rich liquid before the distillation column. In this way, a large fraction of glycerol is prevented from entering the column reboiler, otherwise, the high viscosity of the glycerol and its high temperature exposure would cause difficulties in the separation process in terms of both cost and efficiency. Therefore, the distillation column is mainly used for the methanol–water separation after the flash vapor streams have been concentrated and mixed. [13]

This study should be considered a preliminary design rather than a fully detailed plant design. In the preliminary calculations, ideal K-values have been assumed, it has also been assumed that glycerol is almost non-volatile and that the relative volatilities of methanol and water are constant. These simplifications ensure consistent consideration of the main separation strategy; equipment loads and cost-efficiency trade-offs. For a more detailed design, a thorough examination of non-ideal vapor-liquid equilibrium modelling, such as NRTL or UNIQUAC, and physical property data validated within the actual operating range are required. [6–8, 15]

2. Design Basis and Main Assumptions

The design is based on the project requirements and composition data. The total feed flow rate is 100 kmol/h, and the feed consists entirely of methanol, water and glycerol. The target purity of each recovered product stream is at least 95 percent on a molar basis. It has been assumed that all units operate in a steady state throughout this recovery process and that no chemical reactions take place in the separation section. [1,2]

The two flash drums have been modelled as equilibrium stages. The K values for methanol and water have been estimated using Raoult's law calculations based on Antoine vapor pressure data. However, as glycerol is much less volatile than methanol and water under the selected conditions, it has been treated as a practically non-volatile substance. While this assumption is reasonable enough for preliminary analysis, it remains a significant limitation as the final glycerol purity is close to the specified minimum value. [6–8,15]

It was assumed that the vapor streams from Flash 1 and Flash 2 were fully condensed and feed-conditioned before entering the distillation column. Thus, the column feed could be treated as a saturated liquid, and a value of 1 was used for q in the McCabe-Thiele calculations. For the column calculation, methanol was selected as the light component and water as the heavy component. A constant relative volatility

value of 4.5 (alpha) was used as an average default value for the methanol-water system. A general plate efficiency of 70% was used to convert the theoretical stages into actual plates. [4]

Vapor pressure calculations for methanol and water, using the Antoine parameters from the NIST.

Table 1. Design Basis and Main Assumptions

Design Object	Selected Basis
Feed flow rate	100 kmol/h
Feed composition	12.92 mol% MeOH, 61.18 mol% water, 25.90 mol% glycerol
Allowed separation units	Two flash drums and one continuous multistage distillation column
Target products	Minimum 95 mol % target component in each recovered product
Main thermodynamic basis	Preliminary ideal K-values; constant alpha for MeOH-water column
Thermodynamic Model	$K_i = P_i^{sat}/P$ for methanol and water preliminary VLE screening, $\alpha=4.5$ constant relative volatility for the methanol-water
Glycerol volatility	nearly nonvolatile because of glycerol has much lower volatility than methanol and water
Column feed condition	Condensed flash vapors, treated as saturated liquid ($q = 1$)
Column model	Binary methanol–water separation with trace glycerol
Column pressure	1.013 bar abs
Column type	Sieve-plate column
Reflux selection	$R = 1.20R_{min}$
Tray efficiency	70% overall efficiency for preliminary design

Chemistry WebBook were determined as a result of academic research. NIST gives the Antoine form as $\log_{10}(P) = A - B/(T + C)$, where P is vapor pressure and T is temperature. Glycerol molecular weight was taken as 92.0938 kg/kmol from NIST. Taking into account the information provided in the lecture on the project and the approaches outlined in Chapter 6 as lecture notes: $A_n = A_c - A_d$, $A_a = A_c - 2A_d$, $A_d/A_c = f(L_w/D_c)$, in order; the gas velocity based on the overflow ratio, borehole area boundaries, friction control, dry pressure drop, ventilation pressure drop, leakage control and downpipe backup calculations have been provided. These have been incorporated into the hydraulic section. As no cost proposal data from any material supplier or cost basis specified by the lecturer was provided, the cost model has been kept at a preliminary review level rather than a comprehensive analysis. [3–7, 11, 12]

3. Selected Process Configuration

The following process flow was selected for the process configuration: first, two flash drums, followed by a single continuous multi-stage distillation column. The material enters Flash Drum 1, which operates at 90 °C and 0.50 bar absolute pressure; here, a vapor stream rich in methanol and water is separated from the heavier liquid phase. The liquid contains almost all the glycerol and leaving Flash Drum 1 then enters Flash Drum 2. Flash Drum 2 operates at 120 °C and 0.08 bar absolute pressure and, under vacuum conditions, removes the remaining methanol and water from

the glycerol-rich liquid. The liquid product exiting Flash Drum 2 is recovered as a glycerol-rich product stream. The vapor streams from both flash drums are fully condensed in a pre-column condenser/mixer, pumped and conditioned by the column feed heater, and then fed into a distillation column operating at approximately atmospheric pressure. In the column, methanol is recovered as the distillate product from the top of the column, and water is recovered as the bottom product from the bottom of the column. [1,13]

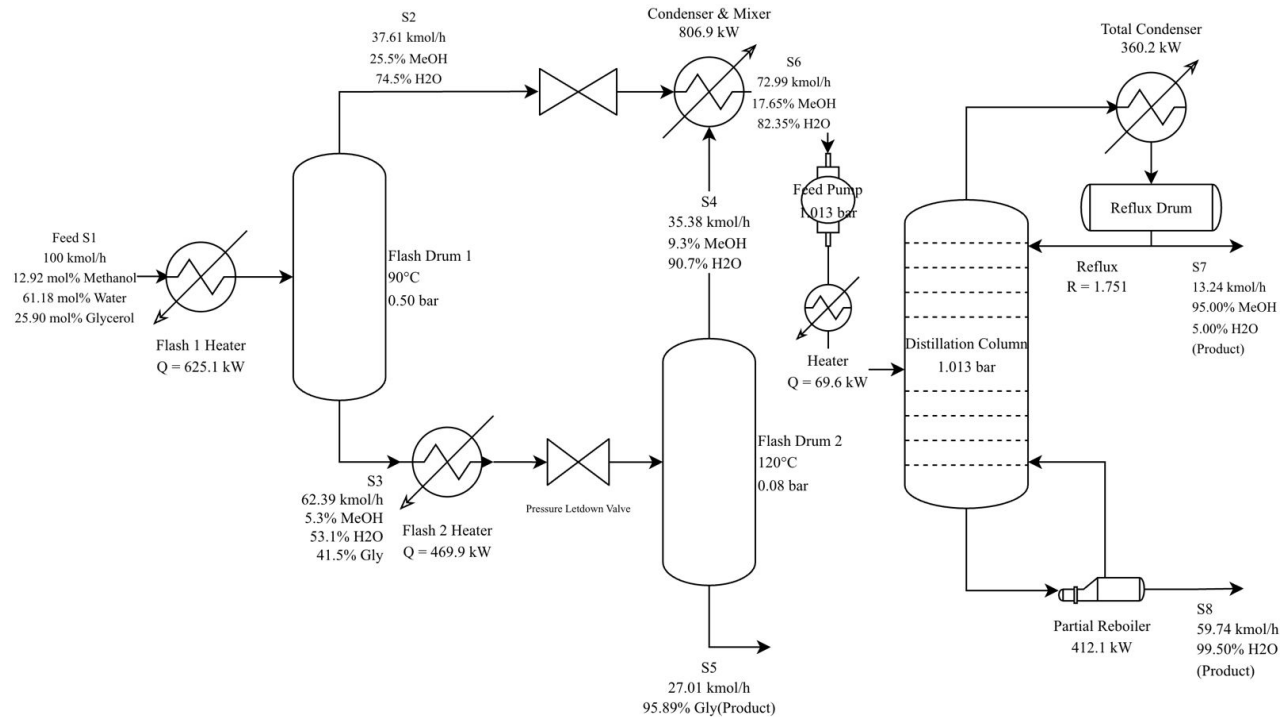


Figure 1. Simplified Block Flowsheet of the Selected Recovery Process

This configuration is used because the difference in volatility between methanol, water and glycerol is quite significant. Methanol is the most volatile component and therefore enters the vapor phase first; whilst water has moderate volatility, glycerol is the least volatile component and remains predominantly in the liquid phase. By using two flash drums upstream of the distillation column, the glycerol in the liquid product is concentrated and the amount of glycerol entering the column is significantly reduced. Thus, the configuration that uses two flash distillation units becomes a more advantageous option than feeding the initial three-component mixture directly into the distillation column. This is because feeding directly into the column carries large quantities of glycerol, which has a high boiling point, into the reboiler section, increasing the liquid viscosity, the risk of overheating in the reboiler and energy consumption. As a result, the selected order increases both separation efficiency and process safety, whilst allowing the distillation column to be designed and developed primarily as a methanol–water separation unit. [1,8,13,15]

4. Overall Mass Balance

4.1 Feed Stream

Table 2. Feed Stream Compositions and Mass Flow

Component	Mole fraction	Flow (kmol/h)	MW (kg/kmol)	Mass flow (kg/h)
Methanol	0.1292	12.92	32.0419	413.98
Water	0.6118	61.18	18.0153	1102.17
Glycerol	0.2590	25.90	92.0938	2385.23
Total	1.0000	100.00	—	3901.38

Although water is the largest component by molar fraction, glycerol has the highest mass flow rate, as its molecular weight is much higher than that of methanol and water. This is one of the advantages of reducing the flow of glycerol entering the distillation column, as it is beneficial in terms of both hydraulic design and energy consumption. [6-8]

4.2 Stream Summary

Table 3. Overall Process Stream Summary

Stream	Total kmol/h	MeOH kmol/h	Water kmol/h	Glycerol kmol/h	MeOH mol%	Water mol%	Glycerol mol%
S1 Feed	100.00	12.920	61.180	25.900	12.92	61.18	25.90
S2 Flash 1 vapor	37.61	9.587	28.025	0.000016	25.49	74.51	~0.00004
S3 Flash 1 liquid	62.39	3.333	33.155	25.900	5.34	53.14	41.51
S4 Flash 2 vapor	35.38	3.300	32.079	0.000034	9.33	90.67	~0.00010
S5 Glycerol product	27.01	0.034	1.076	25.900	0.12	3.99	95.89
S6 Column feed	72.99	12.886	60.104	0.000050	17.65	82.34	~0.00007
S7 Methanol product	13.25	12.588	0.663	0.000	95.00	5.00	0.00
S8 Water product	59.74	0.299	59.441	0.000050	0.50	99.50	~0.00008

The material balance shows that all three product streams achieve the required purity criteria. The methanol product contains 95.00% molar methanol, the water product approximately 99.50% molar water, and the Flash 2 liquid reaches a value of 95.89% molar glycerol. The glycerol product exceeds the target value, but the difference is not significant, therefore it can be considered that the Flash 2 pressure and temperature act as critical design variables affecting the purity level. [1,2]

5. Flash Distillation Design

Each flash drum is modeled as one equilibrium stage. The total and component material balances are:

$$F = V + L$$
$$Fz_i = Vy_i + Lx_i$$

The equilibrium relation is:

$$y_i = K_i x_i$$

For methanol and water:

$$K_i = \frac{p_i^{sat}}{P}$$

The vapor fraction is calculated using the Rachford–Rice equation:

$$\sum_i \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0$$

where:

$$\beta = \frac{V}{F}$$

[6-8,15]

5.1 Flash 1

Flash Drum 1 operates at 90°C and 0.50 bar absolute pressure. At this condition, the vapor fraction, (β) is 0.3761, giving 37.61 kmol/h vapor and 62.39 kmol/h liquid. The vapor contains about 25.49 mol% methanol and 74.51 mol% water, while glycerol remains almost completely in the liquid stream. Flash 1 therefore reduces the volatile load in the glycerol-rich liquid before it enters the second flash drum. [6,7]

5.2 Flash 2

Flash Drum 2 operates at 120°C and 0.08 bar absolute pressure. Vacuum operation is required because glycerol begins to degrade thermally above approximately 200°C; lowering the pressure to 0.08 bar reduces the bubble points of methanol and water to well below 50°C, so that at 120°C both components exhibit K-values above 24 ($K_{\text{MeOH}} \approx 75$, $K_{\text{water}} \approx 25$) and are effectively stripped from the glycerol-rich liquid without exceeding the glycerol thermal degradation limit. The 0.08 bar operating pressure was selected as the lowest practical level for a liquid-ring vacuum pump and achieves the required ≥ 95 mol% glycerol purity targeted. The corrected vapor fraction is 0.5671, producing 35.38 kmol/h of vapor and 27.01 kmol/h of liquid. The Flash 2 liquid contains 25.90 kmol/h of glycerol within the total liquid flow of 27.01 kmol/h; this represents 95.89 mol% glycerol. This provides the minimum glycerol purity specification; however, if the process is further developed, the design should be validated using a more advanced VLE model. [8,15]

6. Distillation Column Design

The vapor streams from Flash 1 and Flash 2 are mixed after the condensation process and fed into the distillation column. As the flash unit removes almost all the glycerol, the mixture fed into the column contains only trace amounts of glycerol and can be modelled as a methanol–water

binary mixture for preliminary design. Methanol is selected as the light component and water as the heavy component. [4]

The selected product specifications are x_D , MeOH=0.950 for the distillate and x_B , MeOH=0.005 for the bottom product. The column feed rate is z_F , MeOH=0.1765 and z_F , water=0.8235. In the preliminary calculation, a constant relative volatility ratio of 4.5 was used for methanol-water. The minimum theoretical number of stages was estimated using the Fenske equation: [4,15]

$$N_{min} = \frac{\ln \left[\frac{x_{D,LK} x_{B,HK}}{x_{B,LK} x_{D,HK}} \right]}{\ln \alpha}$$

The minimum reflux ratio was estimated using a binary pinch approach, and the actual number of theoretical stages was estimated using a Gilliland-type correlation. A reflux multiplier of 1.20 was selected ($R/R_{min} = 1.20$) because it gave the lowest screening total annualized cost among the tested cases. The resulting column design is summarized below. [4]

Table 4. Distillation Column Design Results

Column results	Value
Column feed	72.990 kmol/h
Feed methanol mole fraction	0.1765
Distillate flow rate	13.250 kmol/h
Bottoms flow rate	59.740 kmol/h
Methanol in distillate	95.00 mol%
Water in bottoms	99.50 mol%
Minimum reflux ratio, R_{min}	1.459
Selected R/R_{min}	1.20
Selected reflux ratio, R	1.751
Minimum theoretical stages	5.48
Theoretical stages	13.31
Rounded theoretical stages	14
Overall tray efficiency	0.70
Actual trays	20
Estimated feed tray	Tray 11 from top
Column pressure	1.013 bar absolute pressure
Estimated top temperature	Approximately 66–70°C
Estimated bottom temperature	Approximately 100°C

While the distillate fully meets the specification for 95% mol methanol, the bottom product exceeds the specification for 95% mol water. As the column is primarily designed for methanol-water separation, it is assumed that the trace amounts of glycerol entering the column have a negligible effect on the pre-column equilibrium. [1]

7. Product Purity and Recovery

Product purity and component recovery were calculated on a molar basis. The general expressions used are:

Purity is calculated as:

$$\text{Purity of product } j = \frac{\text{moles of desired component in product } j}{\text{total moles of product } j}$$

Recovery is calculated as:

$$\text{Recovery of component } i = \frac{\text{moles of component } i \text{ in desired product}}{\text{moles of component } i \text{ in feed}} \times 100$$

For example, the glycerol product purity is calculated as $25.900/27.01 = 0.9589$, or 95.89 mol%. The methanol recovery is calculated as $12.588/12.920 = 0.9743$, or 97.43%. The calculated product purities and recoveries are listed below.

Table 5. Product Purity and Recovery Results

Product stream	Target component	Product purity	Recovery
Distillate	Methanol	95.00 mol%	97.43%
Column bottoms	Water	99.50 mol%	97.16%
Flash 2 liquid	Glycerol	95.89 mol%	≈100%

These results indicate that the selected process configuration is applicable at the preliminary design level. Methanol and water recoveries are both above 97%, while glycerol remains almost completely in the liquid product because of its very low volatility relative to methanol and water.

8. Sieve-Plate Column Internal Design

The internal design of the sieve-plate column was carried out using the plate design procedure in Section 6. The rectifying and stripping sections were screened separately because the vapor and liquid traffic are different above and below the feed tray. Although the stripping section has the higher liquid flow rate, the rectifying section controls the final diameter under the selected property assumptions because it gives the larger vapor-side diameter requirement. The calculation began with the vapour and liquid mass flow rates, followed by preliminary estimates for vapour density, liquid density and surface tension. The plate layout was selected by specifying the weir length ratio (L_w/D_c), downcomer area fraction (A_d/A_c), hole diameter (d_h), hole pitch (PT), weir height (h_w), and hole area fraction before performing the hydraulic checks. Based on these values, the column diameter was calculated from the vapour capacity and subsequently checked against criteria for overflow, leakage, pressure drop, entrainment, downcomer backup and downcomer residence time. As approximate physical properties were used, the results should be interpreted as a screening-level plate design rather than a final mechanical design. [3,5,12,15]

The net area is:

$$A_n = A_c - A_d$$

Since:

$$a = \frac{A_d}{A_c}$$

then:

$$A_n = A_c(1 - a)$$

The vapor mass flow relation is:

$$\dot{G} = \rho_G u_G A_n$$

The column cross-sectional area is:

$$A_c = \frac{\pi D_c^2}{4}$$

The flooding velocity is calculated as:

$$u_{GF} = K_{1c} \sqrt{\frac{\rho_L - \rho_G}{\rho_G}}$$

where:

$$K_{1c} = K_1 f \left(\frac{\sigma}{0.02} \right)^{0.2}$$

The operating gas velocity is selected as a fraction of the flooding velocity:

$$u_G = 0.6 - 0.8 u_{GF}$$

The course notes also state that the operating gas velocity should be taken as a percentage of flooding velocity and that K_1 is obtained from the capacity factor chart using the flow parameter. The operating gas velocity was selected as a fraction of the flooding velocity, and the final column diameter was rounded to a practical larger value.

The selected operating gas velocity was taken as a fraction of the flooding velocity. The hole area, hole velocity, weir geometry, pressure drop, downcomer backup, and entrainment were then checked using the course handout method. The most important hydraulic results are summarized below. [3,5]

Table 6. Sieve-Plate Column Hydraulic Design Summary

Hydraulic items	Value	Criterion
Design approach	Rectifying and stripping sections screened	Larger diameter selected
Rectifying vapor mass flow	0.317 kg/s	From column boilup
Stripping vapor mass flow	0.183 kg/s	From column boilup

Rectifying required diameter	0.685 m	From vapor capacity
Stripping required diameter	0.458 m	From vapor capacity
Selected column diameter	0.70 m	Standardized larger size
Rectifying flooding fraction	67.0%	Below 70% design limit
Stripping flooding fraction	30.0%	Below 70% design limit
Plate spacing	0.45 m	Preliminary tray spacing
Hole diameter	5 mm	Common sieve-tray value
Hole area fraction	0.05	Within 0.05–0.15
Plate layout parameters	$L_w/D_c = 0.715$, $A_d/A_c = 0.0945$, $d_h = 5$ mm, $PT = 21.30$ mm, $h_w = 50$ mm	Course step 4 layout selection
Weeping check	Pass	$u_h = 19.82$ m/s $>$ $u_{hm} = 15.41$ m/s
Total gas pressure drop	56.50 mm liquid	Acceptable screening value
Downcomer backup	185.70 mm	Less than 225 mm limit
Downcomer residence time	32.17 s	Greater than 5 s
Estimated entrainment	0.012	Less than 0.1
Final controlling section	Rectifying section	Controls diameter
Design status	Satisfied	Preliminary hydraulic checks passed

The rectifying section requires a diameter of 0.685 m, while the stripping section requires 0.458 m under the screening assumptions. Therefore, the selected standard column diameter was rounded to 0.70 m. This gives an actual flooding fraction of 67.0% in the rectifying section and 30.0% in the stripping section, both below the 70% design limit. In the detailed hydraulic check, the downcomer backup is 185.70 mm, which is lower than the permitted value of 225 mm for a 0.45 m plate spacing. The downcomer residence time is 32.17 seconds, which is greater than the recommended minimum value of 5 seconds. Thus, the selected plate design passes the basic preliminary hydraulic checks. [3,5]

9. Energy Balance

Heat losses to the surroundings were neglected in the preliminary energy balance. The flash heater duties were estimated by including both sensible heating and partial vaporization, while the column reboiler and condenser duties were estimated mainly from latent heat requirements. The following simplified expressions were used: [11,12,15,16]

$$Q_{Sensible} = n C_p \Delta T$$

$$Q_{latent} = n_{vaporized} \Delta H_{vap}$$

$$Q_{total} = Q_{sensible} + Q_{latent}$$

$$Q_{reboiler} \approx V_{boilup} \Delta H_{vap}$$

$$Q_{condenser} \approx V_{top} \Delta H_{vap}$$

Table 7. Energy Balance Summary

Duty Items	Sensible Duty (kW)	Latent/Condensing Duty (kW)	Total Duty (kW)
Flash 1 Heater	214.3	410.8	625.1
Flash 2 Heater	74.9	395.0	469.9
Column Feed Heater	69.6	0	69.9
Column Reboiler	0.0	411.9	411.9
Column Condenser	0.0	360.2	360.2
Pre-Column Condenser	0.0	805.9	805.9
Total Cooling Duty	-	-	1166.05
Total Heating Duty	-	-	1576.5

The largest heating requirements relate to the two flash heaters. Flash 1 requires significant sensible heating from the feed condition to 90°C in addition to vaporization duty. Flash 2 requires less sensible heating because its feed has already been heated in Flash 1, but it still requires considerable vaporization duty to remove methanol and water from the glycerol-rich liquid. To ensure the energy balance is complete, the column feed heater and pre-column condenser loads are also reported. However, according to the cost scope note prepared by the instructor, although flash heater loads and pre-column heat exchanger/condenser loads are calculated, they are not included in the TAC calculation.

10. Preliminary Cost Model

The main cost equations and cost functions used in the screening calculation were:

$$\text{Column shell cost} = 60000(D/0.5)^{1.1}(H/10)^{0.8}$$

$$\text{Sieve tray cost} = 1200N_{trays}(D/0.5)^{1.2}$$

$$\text{Flash drum cost} = 18000(V_{vessel}/1)^{0.6}$$

$$\text{Heat exchanger cost} = 15000(Q/100)^{0.65}$$

$$\text{Installed capital} = \text{Bare equipment cost} \times \text{installation factor}$$

$$\text{Annualized capital} = \text{Installed capital} \times \text{capital recovery factor}$$

$$\text{Steam cost} = Q_{heating} \times \text{operating hours} \times 3.6 \times 10^{-3} \times \text{steam cost per GJ}$$

$$\text{Cooling cost} = Q_{cooling} \times \text{operating hours} \times 3.6 \times 10^{-3} \times \text{cooling cost per GJ}$$

$$\text{TAC} = \text{Annualized capital} + \text{included steam utility cost}$$

$$+ \text{included cooling utility cost} + \text{pump electricity cost}$$

The cost basis used in the Excel file was 8000 h/year operation, steam at 18 \$/GJ, cooling utility at 0.4 \$/GJ, electricity at 0.12 \$/kWh, an installed-cost factor of 2.7, and a capital recovery factor of 0.20 year⁻¹. [11,12,16]

Table 8. Preliminary Cost Model Summary

Cost Items	Value
Column Shell	\$97,152
Sieve Trays	\$35,939
Flash Drum 1	\$18,000
Flash Drum 2	\$13,248
Vacuum + Liquid Feed Pump	\$40,000
Column Reboiler + Total Condenser	\$72,144
Bare Equipment Subtotal	\$276,483
Installed Capital	\$746,503
Annualized Capital	\$149,301/year
Steam Utility	\$213,508/year
Cooling Utility	\$4,149/year
Pump Electricity	\$19,256/year
Estimated Total Annualized Cost	\$386,213/year

An annual total cost model was used to compare design alternatives. The heat duties of Flash Drum 1, Flash Drum 2, and all pre-column heat exchangers/condensers were calculated and reported in the energy balance, but they were not included as utility costs in the TAC calculation. Therefore, the reported TAC represents the distillation-column cost basis with flash drum vessels and required pumps included as capital screening items. [11,12,16]

The largest costed operating contribution is the column reboiler steam duty, while the pre-column heating and cooling duties remain outside the TAC by instruction. As the cost model is at a screening level and sensitive to assumed service prices, the final TAC has been reported as approximately \$0.386 million per year. [11,12,16]

11. Reflux Ratio Sensitivity as a Control Parameter

A reflux ratio sensitivity was performed to evaluate the trade-off between column capital cost and energy cost. Increasing the reflux ratio decreases the required number of stages, but it increases vapor and liquid traffic, which raises condenser and reboiler duties. Because the pre-column duties are outside the TAC scope, this comparison was performed on the distillation-column cost basis. The lowest tested reflux multiplier gives the lowest TAC. [11,12,16]

Table 9. Reflux Ratio Sensitivity Results

R/R _{min}	Actual Reflux Ratio	Theoretical Stages	Actual Trays	Reboiler Duty (kW)	TAC (\$/year)
1.20	1.751	13.31	20	412	386,213
1.30	1.897	12.06	19	434	396,471
1.50	2.189	10.49	16	478	414,214
1.80	2.627	9.20	15	543	449,912
2.00	2.919	8.66	13	587	470,002

The selected design uses $R/R_{\min} = 1.20$, corresponding to $R = 1.751$. This case gives the lowest screening TAC among the alternatives considered. The result is consistent with the revised course cost scope, where flash and pre-column heat duties are reported but excluded from TAC. [11,12,16]

12. Equipment List

The main process equipment required for the selected configuration is listed below. Supporting equipment such as pumps, control valves, pressure relief devices, vacuum protection systems and measuring instruments has been included as it is necessary for safe and continuous operation. [1,11,12,16]

Table 10. Major Equipment Used List

Equipment	Quantity	Main Function	Operating/Design Condition
Flash Drum 1	1	Initial methanol-water vapor recovery	90°C, 0.50 bar abs
Flash Drum 2	1	Glycerol concentration	120°C, 0.08 bar abs
Flash 1 Heater	1	Heat feed to Flash 1 condition	625.1 kW
Flash 2 Heater	1	Heat Flash 1 liquid to Flash 2 condition	469.9 kW
Distillation Column	1	Methanol-Water separation	1.013 bar abs, D= 0.7 m
Sieve Trays	20	Vapor-liquid contacting	0.45 m tray spacing
Reboiler	1	Provide column boilup	411.9 kW
Condenser	1	Condense top vapor	360.2 kW
Pre-Column Condenser	1	Condense S2+S4 flash vapors before column feed	805.9 kW
Column Feed Heater	1	Bring S6 to saturated liquid	69.6 kW
Reflux Drum	1	Collect condensate and provide reflux	Atmospheric top section
Pumps	As needed	Transfer liquid products and maintain vacuum	Vacuum pump Flash 2 and column feed pump
Relief and Vacuum Protection	As needed	Overpressure and vacuum protection	All pressure/vacuum vessels
Instrumentation	As needed	Pressure, temperature, level, and flow control	All units

13. Safety Considerations

The use of methanol in this design is the most significant safety concern. Methanol is both flammable and toxic. OSHA classifies methanol as a flammable liquid with a flash point of 52°F and flammable limits in air ranging from 6.0% to 36% by volume. Therefore, equipment handling methanol-containing streams must be enclosed, earthed, bonded and adequately ventilated. Electrical equipment in the vicinity of methanol vapour must be suitable for flammable atmospheres, and vapour leaks must be minimised. Monitoring of the lower explosion limit is recommended in enclosed areas where methanol vapour may accumulate. [9,10]

Flash drums, particularly Flash Drum 2 at 0.08 bar abs pressure, operate at low pressure. Operation under vacuum carries the risk of air ingress. If air enters equipment containing methanol

vapour, a flammable methanol-air mixture may form. Therefore, the vacuum system must include air leak detection, adequate ventilation, pressure control and vacuum protection. Flash drums and columns must also be equipped with venting devices, level control, temperature control, pressure alarms and emergency shutdown. [9,10]

Reboiler overheating should be avoided because even trace glycerol-containing streams may become thermally sensitive at elevated temperatures. Although the column feed contains only trace glycerol, conservative reboiler temperature control is still recommended. The selected flash-first configuration improves process safety by removing most glycerol before distillation and reducing the chance that a viscous, high-boiling glycerol-rich liquid enters the column reboiler. Operators should use appropriate gloves, eye protection, adequate ventilation, and controlled or closed-loop sampling systems when working near methanol-containing equipment. [8,15]

14. Reuse Applications

The possible reuse applications of the recovered streams are summarized below:

Table 11. Reuse Applications of Recovered Products

Recovered stream	Purity / recovery	Potential reuse
Methanol product	95.00 mol%, 97.43% methanol recovery	Recycle to biodiesel transesterification after water-content check
Water product	99.50 mol%, 97.16% water recovery	Process wash water or utility reuse if residual methanol limit is satisfied
Glycerol product	95.89 mol%, approximately 100% glycerol recovery	Crude glycerol by-product or feedstock for further purification

The recovered methanol stream has the highest direct process value because methanol is normally used in excess in biodiesel transesterification. Recycling the 95 mol% methanol product can reduce fresh methanol demand, although its water content should be checked before recycle because water may affect reaction performance and catalyst behavior. [13] The water bottoms stream is about 99.50 mol% water and may be reused as process wash water or in utility-related service if the residual methanol concentration satisfies site requirements. This reuse option supports the circular economy objective by reducing freshwater demand and lowering wastewater generation. [1,13] The glycerol-rich liquid contains 95.89 mol% glycerol and can be considered a crude glycerol product. This stream may be sold or used as feedstock for further purification. Higher-value applications, such as pharmaceutical or food-grade glycerol, would require additional polishing steps outside the equipment scope allowed in this project. [14]

15. Conclusion

In conclusion, an applicable preliminary recovery system was developed for the methanol-water-glycerol feed considered in this study. The selected process uses two flash drums followed by one continuous sieve-plate distillation column, satisfying the project restriction on available separation units. Flash Drum 1 operates at 90°C and 0.50 bar absolute pressure to remove a methanol-water vapor stream, while Flash Drum 2 operates at 120°C and 0.08 bar absolute pressure to concentrate the glycerol-rich liquid. The Flash 2 liquid product reaches 95.89 mol% glycerol. [1] The combined methanol-water stream from the flash section is separated in an atmospheric distillation column. The selected column produces a 95.00 mol% methanol distillate and a 99.50 mol% water bottoms product. The design requires approximately 14 theoretical stages, corresponding to 20 actual trays at 70% tray efficiency. The selected reflux ratio is $R = 1.751$, based on $R/R_{\min} = 1.20$, because this case gives the lowest screening total annualized cost among the reflux ratios evaluated. [4,5] The sieve-plate hydraulic design gives a selected column diameter of 0.70 m and a tray spacing of 0.45 m. The estimated total heating duty is approximately 1.58 MW, while the preliminary total annualized cost is approximately \$0.386 million/year under the instructor-defined cost scope. The main limitation of the design is the simplified thermodynamic model, especially the use of ideal K-values and constant relative volatility. Therefore, a comprehensive final design should use a nonideal activity-coefficient model and verified physical-property data. Within the assumptions of this preliminary study, the proposed configuration satisfies the required purity targets and provides a reasonable balance between separation performance, safety, and cost. [11,12,15,16]

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